# Sahaj Bhadja

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## Summary

Software engineer and 2026 MSCS candidate focused on **C++**, **data structures**, **low-level systems**, Linux-style debugging, testing, and performance-minded backend tools for trading-adjacent environments.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *LA, California*
- Built Azure-backed AI education platform with normalized SQL schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across releases.
- Reworked FastAPI background jobs with explicit states, failure isolation, and SQL-backed status tracking, processing 1K+ AI jobs while tripling throughput under concurrent instructor workloads, tickets, and reviews.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *LA, California*
- Designed concurrent file-ingestion services with ThreadPoolExecutor, OCR, and multimodal AI review, reducing feedback turnaround 60% while keeping batch processing testable, observable, and responsive for instructors and QA.
- Implemented production grading workflows for uploads, rubrics, history, and model selection, supporting 150+ instructors and 10K+ daily requests through REST APIs, dashboards, bug fixes, and releases reliably.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles*
- Built a Decision Transformer for autonomous-driving offline RL, improving planning 18% in simulation while processing 100K+ trajectory logs through reproducible Python evaluation pipelines for systems research experiments.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% across simulated baselines for performance analysis workflows consistently.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* *India*
- Troubleshot large-scale API integrations using OOP patterns, SQL workflows, AWS Glue, and S3, improving response times 30% while reducing technical debt 20% across release cycles and defects.
- Engineered cross-platform microservices with Jenkins CI/CD, VM automation, and 20+ PyTest, Mockito, unit, and integration suites while tracking defects through Agile sprint delivery with QA teammates consistently.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*
- Built backend APIs for booking, slot selection, reservation tracking, coupons, and payment-status updates across 50K+ records, delivering sub-200-millisecond responses for client-facing workflows and support reviews reliably remotely.
- Mapped 12+ integration points across Flask APIs, Firebase synchronization, webhooks, payments, and cross-platform delivery, reducing design errors 25% through early stakeholder architecture reviews and documentation handoffs collaboratively.

## Skills

**C++ & Systems:** C++, STL, Boost, Data Structures, Algorithms, OOP, Memory Management, Multithreading, Concurrency, Linux, Shell, Debugging
**Trading & Performance:** Low Latency, Market Data, Order Books, Research Tools, Resource Utilization, Performance Profiling, Refactoring, Bug Fixes
**Backend & Data:** Python, Java, SQL, PostgreSQL, SQL Server, FastAPI, Flask, REST APIs, Webhooks, Redis, AWS, S3, Docker
**Testing & Delivery:** Unit Testing, Integration Testing, PyTest, Mockito, CI/CD, Jenkins, GitHub Actions, Git, Agile/Scrum, Code Reviews, Technical Communication

## Projects

**Market Data Replay Engine** | *C++, STL, Boost, Linux, Order Books, Git*
- Designed a C++ trading-systems concept with STL containers, Boost-style utilities, market-data replay, order-book state updates, latency instrumentation, Linux deployment assumptions, unit tests, and debug traces for performance reviews.

**CPU-scheduling Simulator** Live | *C++, React, JavaScript, EJ-2*
- Implemented React controls and C++ scheduling logic for FCFS, SJF, SRT, RR, priority, and MLQ algorithms, calculating turnaround metrics across 1K+ process combinations with dynamic visualization updates.

**FinNexus** Live | *Next.js, TypeScript, MongoDB, Plaid, WebSockets*
- Built a fintech platform supporting secure bank linking, real-time transaction updates, tokenized access, and filtering across 1M+ transactions, using TypeScript, MongoDB, Plaid, WebSockets, and deployment automation reliably.